Om Singh Chauhan

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Skills

- **Languages**: Python, Java, C/C++, JavaScript
- **Libraries/Frameworks**: Scikit-learn, Pandas, Matplotlib/Seaborn, PyTorch, Flask, Next.js, Vue.js
- **Tools/Platforms**: MySQL, Vuex, MS-Excel, Power BI, Git/GitHub
- **Soft Skills**: Collaboration, Adaptability, Problem-Solving, Rapid Learning, Time Management

Projects

Intelligent CPU Scheduler – Predictive Scheduling Simulator | *GitHub* Apr' 25

- Developed a comprehensive simulation framework that provides real-time heuristic suggestions for optimal process execution, surpassing standard deterministic models.
- Engineered a Large Sample Testing module to generate stochastic datasets, validating a 91% prediction accuracy across diverse workload simulations.
- Designed an interactive interface with Matplotlib and Tkinter to visualize dynamic Gantt charts and compare metrics like CPU utilization and response times.
- **Tech:** Python, NumPy, Matplotlib, Tkinter, Pandas

Learn-PGT – AI Powered Learning Assistant | *GitHub* Mar' 25

- Built an intelligent education platform that leverages Generative AI to transform static course names into structured learning modules.
- Integrated Google Gemini API to engineer a Weakness-Based Quiz system that dynamically generates questions for targeted revision.
- Implemented server-side logic using Node.js and Express to handle API requests and validate user inputs before processing AI prompts.
- **Tech:** Next.js, Node.js, Express, Google Gemini AI.

Training

Python for Machine Learning/AI | *CSE Pathshala* Jun' 25 - Jul' 25

- Gained in-depth theoretical knowledge of ML algorithms and applied them practically using Scikit-learn for **feature engineering** and **predictive modeling**.
- Collaborated on the JFK Airport Weather Analysis project, specifically managing **outlier detection** to ensure data integrity.
- Optimized model parameters with the **Elbow Method**, determining the ideal configuration to minimize error.

Certificates

- Introduction to Machine Learning - Elite | *NPTEL* Oct' 25
- The Bits and Bytes of Computer Networking | *Google* Sep' 24
- Introduction to Hardware and Operating Systems | *IBM* Sep' 24

Achievements

- Filed a Patent for an AI-driven Space Planner & Real-Time Crew Health Logger Apr' 25
- Won first place in an intra-departmental English comprehension competition focused on literary analysis Sep' 24
- Achieved 5-star Gold Rating in Python | *HackerRank* Nov' 23
- Scaled a personal YouTube channel to 1.6 million+ views and 4k+ subscribers. Oct' 22

Education

- **Lovely Professional University** Punjab, India
  Bachelor of Technology - Computer Science and Engineering; CGPA: 9.16 Aug' 23 - Present
- **Indian Institute of Technology (IIT) Madras** Remote
  BS Degree Program - Diploma in Programming; CGPA: 8.0 Aug' 23 - Present
- **Kendriya Vidyalaya No.1 Jalandhar Cantt** Punjab, India
  Intermediate; Percentage: 91 Apr' 22 - Mar' 23
- **Kendriya Vidyalaya No.1 Jalandhar Cantt** Punjab, India
  Matriculation; Percentage: 94.6 Apr' 20 - Mar' 21